Enhancing Motivation in Software Engineering Education through Gamified Agile Project-based Learning

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Abstract

Project-based learning (PBL), e.g., student software development projects, is an essential part of today's Software Engineering (SE) education. They allow students to work on real-world projects and gain practical experience as a team. However, several challenges arise in such projects, including learning new technologies and dealing with communication and coordination issues within the team. These factors can lead to a lack of motivation to contribute to the project and a decrease in productivity, potentially resulting in an insufficient project outcome. This paper aims to promote student motivation in PBL and increase team productivity by applying gamification. We conducted a user and requirements analysis to identify the needs of students and supervisors of such projects. Based on the insights, we designed and implemented DinoDev, a gamified project management tool that combines project management features with gamification elements. The DinoDev concept was evaluated in a student project, indicating increased motivation and team productivity. The findings are valuable for advancing research on using gamification in PBL and for lecturers to improve their students' motivation and team productivity in SE education.

CCS Concepts

• Social and professional topics → Computer science education; **• Human-centered computing → Collaborative and social computing systems and tools**; **• Applied computing → Interactive learning environments**.

Keywords

Project-Based Learning, Student Motivation, Gamification, Scrum, Software Engineering Education

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1 Introduction

Software Engineering (SE) education plays a crucial role in qualifying future software engineers, which makes effective and relevant delivery of SE content in higher education essential [27]. Hereby, special emphasis is placed on developing students' practical skills [9, 42]. Application exercises, projects, and group assignments are indispensable to foster these practical skills, allowing students to interact, collaborate, exchange ideas, and solve problems together [9]. A pedagogical approach supporting this is project-based learning (PBL), where students engage in real-world projects that require applying theoretical knowledge to practical tasks [6]. In SE education, PBL is often applied in "student software development projects", where agile development processes such as *Scrum* [49] are often used to familiarize students with changing customer requirements from diverse stakeholders, working in sprints, and frequent product reviews [42].

However, numerous factors can lead students to lack motivation and engagement [41] to work on these projects. Students often struggle to learn new technologies and usually have no or minimal prior experience in working on more extensive software systems and developing software in larger teams. Further, communication and coordination within the team can be challenging, and an unequal distribution of work can lead to frustration. Low motivation is a critical factor in the failure of software development projects [63], as it leads to poor engagement with tasks, delayed milestones, and suboptimal collaboration. Thus, it is beneficial for all parties involved in these projects to investigate ways to motivate students.

One approach to increase student motivation is gamification, i.e., applying game design elements in non-game contexts [12]. Hence, in this paper, we tackle the stated challenges and introduce the *DinoDev* concept and tool for implementing gamification in student software development projects. Gamification has succeeded in various domains, including education and SE [1, 14, 46]. The combination of gamification with PBL, also referred to as "gamified project-based learning" (GPBL), is a promising approach to improve student learning outcomes [20]. Furthermore, gamification in SE education could be of great importance due to the specificity of SE, which involves frequent, complex, or repetitive tasks [1]. Nevertheless, more empirical research and concepts for the selection of gamified elements are needed [1]. While gamification has been applied to the agile development process Scrum in industry [31], a more focused application in the context of PBL in SE education is still missing. Therefore, our research questions are as follows:

RQ1: *"Does the implementation of gamification enhance student motivation within student development projects?"*

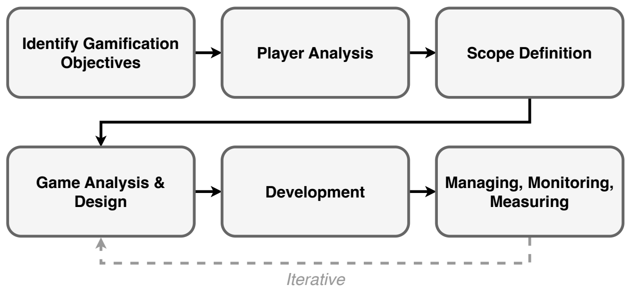

Figure 1: GOAL methodology, adapted from García et al. [15].

> **RQ2:** *"Does the implementation of gamification increase team productivity in student development projects?"*

By answering the research questions, we aim to advance further research in the field of student motivation and gamification in SE education and empower lecturers with a gamification approach to increase student motivation and team productivity in their student software development projects. A demo video of the implemented *DinoDev* prototype is available on YouTube[1].

We followed the methodology depicted in Figure 1 that is based on the *Gamification On Application Lifecycle Management (GOAL)* framework [15]. We conducted (1) an identification of gamification objectives based on literature and a requirements engineering process, (2) a user analysis to collect further requirements for student development projects, followed by (3) the development of our concept and prototype, including both project management features and gamification mechanisms, and (4) an evaluation in a real-world student software development project.

Our evaluation with a project consisting of seven students indicates that the integration of gamification into the agile SE processes is a suitable method to increase student motivation and enhance team productivity in student software development projects.

To summarize, our contributions are: (1) a set of elicited requirements for gamification in PBL courses, (2) an analysis of the supervisors' concerns, perceived benefits, and restrictions, (3) a gamification concept for PBL courses that incorporates gamified agile project management, team-oriented gamification, and student-oriented gamification, and (4) a prototype that implements these concepts, seamlessly integrates Scrum events, and can be used with a variety of traditional issue trackers.

The remainder of this paper is structured as follows: First, Section 2 describes the student software development projects. In Section 3, we discuss related work, followed by presenting the RE process and user analysis in Section 4. Afterward, in Section 5, the *DinoDev* concept is described by considering project management, team- and student-oriented gamification, and Scrum events. Then, Section 6 introduces the developed prototype, followed by evaluating the research questions in Section 7. Finally, after discussing the threats to validity in Section 8 and lessons learned in Section 9, the paper is concluded in Section 10 with an outlook for future work.

[1] https://youtu.be/Fz04QFBUloE

2 Student Software Development Projects

Student software development projects are a form of PBL and are integral to SE education for software engineers due to their practical focus [42]. These projects aim to facilitate the students' ability to develop larger and more complex software systems together as a team, based on the requirements of the "customers" (i.e., the supervisors and examiners). This promotes both the practical and soft skills of the students, and they learn to organize themselves in a team, similar to industrial and open-source real-world software development [39]. At the University of Stuttgart, such student projects are part of both the Bachelor's and Master's degree "Software Engineering" and similar programs. The projects usually consist of development teams of between six and twelve students and typically take place in the fifth semester of the Bachelor's degree and in the second semester of the Master's degree. The Bachelor's project comprises 15 ECTS points (450 hours per student), and the Master's project counts 12 ECTS points (360 hours per student) value. Therefore, we expect that students have already completed courses in programming and SE processes. In these projects, we follow an agile software development method (typically Scrum) with two- or three-week sprints. One student is explicitly the *Product Owner*, and in most cases, another student is the *Scrum Master* of the team and manages the development team and customer communication. Therefore, the students follow typical Scrum events, such as Sprint Plannings, Sprint Reviews, and Sprint Retrospectives.

3 Related Work

Research in the area of gamification suggests that its effectiveness depends on the context in which it is implemented and on the users of the gamified system [17]. Previous studies reveal promising results both in SE and education [1, 14, 46]. However, often only a small selection of simple game elements is used in such studies [16, 29, 40]. Therefore, we identify the need for a more comprehensive gamification concept considering a wider range of game elements and mechanics.

While gamification was utilized in various SE activities, such as software testing [11], requirements engineering [10], code reviews [21, 23], or version control [51], our work aims to gamify the agile SE processes. There are several approaches for a gamified Scrum process: (1) *ScrumHero* [44] is a concept for using gamification for the planning and management of Scrum projects. It includes various reward and scoring mechanisms and replaces the Scrum terminology with playful names such as *wish list* instead of *Product Backlog*. (2) Similarly, Hermanto et al. [18] designed a prototype of a Scrum tool that uses simple game elements, including a user profile and a leaderboard. For both of these approaches, we could not find an empirical validation. (3) In contrast, Marques et al. [31, 32] implemented and evaluated an *Atlassian JIRA* app that uses a variety of game elements with the objective of increasing the team's motivation to apply Scrum practices. This includes, for example, reducing the number of unfinished tasks at the end of the sprint. When applying the app with real-world Scrum teams, they report a slight but statistically insignificant increase in their measured metrics. While their results are somewhat promising, their work is related to an industrial context, which differs from the educational context of this work.

In the educational context, gamification has a positive effect on learning outcomes [4, 28, 47], which also holds for higher education [60] and *e-learning* [48]. We previously conducted an empirical study with 133 SE students, revealing that almost two-thirds believe that gamified elements in Learning Management Systems (LMS) increase student motivation [36]. Gamified LMS focusing on e-learning exist, e.g., the educational platform *IT-REX* [56], tailored to computer science study programs. Based on the concepts from IT-REX, we developed concepts for a gamified intelligent tutoring system called *MEITREX* [34, 35]. It aims to provide an interactive, individual learning experience by incorporating adaptive game elements, personalized feedback, and learning recommendations.

While gamification offers benefits in software engineering education [24, 62], its effects are not as extensively studied or validated as in the broader educational context, underscoring the need for further empirical research [2, 5, 24]. Furthermore, the demand for approaches that go beyond basic game elements also persists in SE education [54]. Such an approach is, for example, *Gamify-IT* [58], a web-based gaming platform for introductory SE courses. The platform provides a computer game-like environment in the form of a role-playing game, where students can play mini-games in a virtual world. Results from a user study with students show that their approach positively affects motivation and engagement. Similarly, Arif et al. [3] developed a web-based learning platform that incorporates gamification to teach programming to vocational high school students, including real-time feedback, mission-based tasks, and competitive game-like elements.

Gamification techniques are mostly applied in SE lectures, but also often in projects (GPBL) [2]. For example, Souza et al. [53] introduced gamified projects to an introductory SE course, resulting in higher student engagement. These projects, however, have a small scope of 60 work hours per student, small team sizes of three to five students, and some of them use fictitious topics, limiting how much these projects represent a real-world scenario. The goal of teaching the Scrum methodology with GPBL has also been investigated in the literature. Sisomboon et al. [52] aimed to enhance engagement and motivation in Scrum-based SE projects by incorporating simple game elements such as points, badges, leaderboards, and challenges, but their approach does not involve the development of a gamified platform or integration with external systems. *Lego* bricks [8, 19], *Miro* boards [13], or the video game *Minecraft* [50] enable replacing the technical aspects of software development with a more playful activity and entirely focus on teaching Scrum. However, we view learning new technologies and improving programming skills as an essential part of these practical development projects, which these approaches neglect.

Besides gamification approaches, SE project management tools for the educational context are sparse. While using industry-standard tools in student development projects may be closer to a real-world setting, specialized tooling can enable supervisors to assess students' learning and identify issues in the project, such as unhealthy work patterns or isolated students in the team. This is achieved by the project management tool *ScrumBoard* [37]. It is specifically designed for SE education and is similar to industry standard tools. We see potential in adding similar student assessment features in *DinoDev* in future implementations.

4 Requirements and User Analysis

Using anonymous online questionnaires, we conducted a quantitative and qualitative analysis of the target users (students and supervisors) and their requirements in student software development projects while identifying prevailing trends that could serve as the foundation for the design of the gamification concept. The questionnaires and all data collected are open source and publicly available on Zenodo[2].

4.1 Requirements Engineering Process

Besides the classic requirements elicitation, requirements analysis, specification, validation, and management [33, 45], we conducted a user analysis of SE students and supervisors to understand them and their needs in the gamification design process [38]. The key questions we aimed to answer with the RE process and the user analysis were as follows: (1) "What are the characteristics of the agile methodology used in student development projects" (2) "Do students feel motivated in their projects?" (3) "What are the requirements for a gamified project management tool?" (4) "Are project supervisors interested in a gamified project management tool?"

Initial requirements were gathered from related work (Section 3) and our perspectives and experiences in supervising student software development projects. From these initial requirements, we derived a questionnaire for the students and supervisors, which can be found on Zenodo[2]. Both questionnaires utilized a range of question formats, such as multiple-choice questions, Likert scales, rankings, and open-ended questions.

The questionnaire for the students consisted of five sections. The first section asked about the agile methodology used in the students' projects. Questions on student motivation and reasons for lack of motivation followed this. The third section assessed the importance of various project management features, including gamification-related elements. Students were then asked about game concepts and elements for a gamified project management tool. Finally, the questionnaire collected demographic data, including assessing the students' player types following *Hexad-12* by Krath et al. [25], a shorter, improved variant of the Hexad scale [61]. These questions allow assessing the player types by Marczewski [30] of a user. As the effectiveness of game elements varies depending on the player types of the users [26, 61], understanding the player types is crucial in the design of gamified systems [15]. The student questionnaire was then distributed to past student software development project participants at the University of Stuttgart and shared in an SE Telegram group and other social media channels. A prerequisite for participation in the questionnaire was that students had completed at least one student software development project.

The questionnaire for the supervisors had a similar structure to the questionnaire for the students, but was shorter and contained mostly open-ended questions. It contained sections dealing with the agile methods used in the students' projects, challenges encountered during these projects, and requirements and concerns regarding the use of a gamified project management tool. We did not collect any additional demographic data on the supervisors. The supervisor questionnaire was then distributed to other project supervisors in various departments of the University of Stuttgart.

[2]https://zenodo.org/doi/10.5281/zenodo.13472406

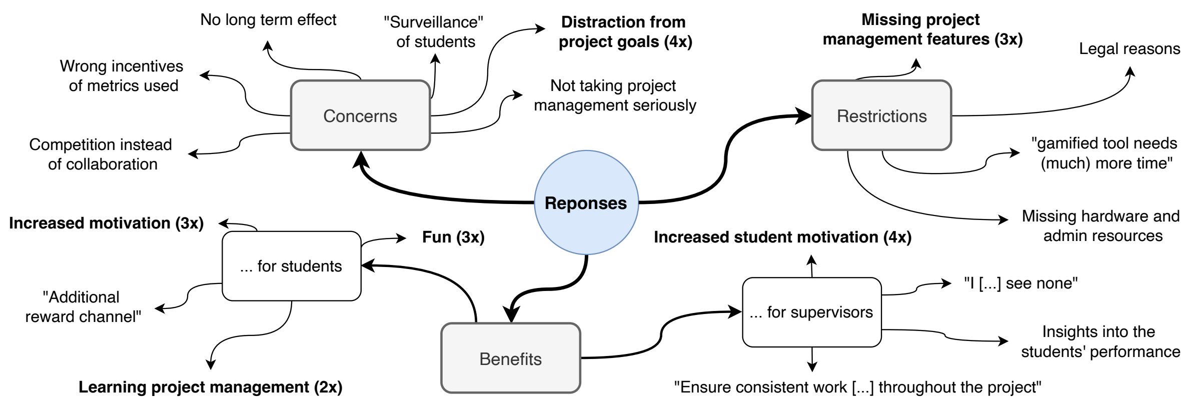

Figure 2: Mind map of the supervisors' concerns, perceived benefits, and restrictions.

4.2 Results and Gathered Requirements

A total of 27 students and seven supervisors took part in the questionnaires. Most of the students identified as male *(89%)* and the average age was 25 *(min. 21, max. 33)*. Further details on the student's demographic details can be found in the data set, but are of no further relevance for the following analysis.

85.2% of the students and all the supervisors stated that the Scrum framework was used in their most recent project, which is consistent with the high adoption rate of Scrum in industry. However, the results show that student projects often slightly deviate from the standard Scrum practices. The most common deviations are introducing sub-teams, having standup meetings less frequently than daily, and omitting the Product Owner role or combining the role with project management tasks.

The survey results show a mixed picture of the students' motivation. While 56% of the students felt more motivated in the project than in regular courses, 76% felt a lack of motivation at some point during the project. Similarly, 86% of the supervisors agreed that students were highly motivated in their projects, while also all of them agreed or strongly agreed that students sometimes lacked motivation. To investigate the suitability of gamification for student software development projects, we asked the students if they think it could positively impact their motivation. 40% of the students agreed with the corresponding statement, while 30% disagreed.

Our survey showed that the most common player type, according to the *HEXAD-12* scores, is the *Philanthropist*, with a median score of ten (out of twelve). In contrast, the *Disruptor* is the lowest player type, with a median score of five. In between are the other types, all with a median score of nine.

All supervisors agreed that they would allow the use of a gamified project management tool in projects under their supervision. 29% even strongly agreed with this statement. All open-question answers of the supervisors are shown as a mind map in Figure 2. Increased motivation of the students was the most mentioned benefit, both for the students and the supervisors. The most mentioned concern was that the tool should not be a distraction from the actual work of the project. Missing project management features was the most mentioned reason that would prevent the supervisors from using the tool.

Based on the responses of the students and supervisors regarding the students' motivation, the requirements for a gamified project management tool have emerged. An essential requirement for supervisors and students is that the implemented tool allows issue and project management, just as existing tools do. Game elements should be secondary to avoid "artificial work". Beyond classical issue management, the tool should offer assistance in planning, daily, and retrospective meetings. Further detailed requirements, such as the presence of an administrator role, have been documented in the related master's thesis [7]. The requirements, as well as the restrictions, benefits, and concerns from Figure 2 are incorporated into the gamification concept in Section 5.

5 Gamification Concept

The proposed gamification concept combines issue and project management features tailored to the needs of student software development projects with gamification features based on the gathered requirements and user analysis in Section 4. Specific measures to address the supervisors' concerns (Figure 2) are reflected in the concept by seamlessly integrating game elements into traditional project management workflows to avoid distraction from project goals. We divide the concept into four parts: (1) Project management features that promote using our developed tool by providing features to support the Scrum process. (2) Team-oriented gamification features that aim to increase the motivation to work collaboratively on the project, strengthen the team, and motivate them to achieve the project goals. (3) Student-oriented gamification features that aim to enhance the motivation of individual students so that they comply with their duties and responsibilities to the project and deliver their work diligently and reliably. (4) Scrum events that are designed to be reduced to the minimum amount of time, increase meeting productivity and effectiveness, and motivate each student to participate actively.

The long-term goal of the *DinoDev* gamification concept is the integration to the MEITREX [35] learning environment. Due to the dinosaur theme of the platform and the underlying IT-REX [56]

Figure 3: The sprint mascot grows with every task completed.

vision of creating a gamified learning environment for computer scientists in the first semesters, the dinosaur theme was adopted for *DinoDev*. Thus, the general idea of *DinoDev* is that the students of the project hatch and raise a dinosaur as a sprint mascot in each sprint (Figure 3). Each task completed by a student contributes to the growth of the dinosaur. If the team completes all planned tasks in a sprint, the dinosaur is fully grown, and the team can start with a new dinosaur in the next sprint. This way, a dinosaur park full of dinosaurs of different species is created over time, each dinosaur representing a sprint. If the team fails a sprint, the dinosaur has to leave the park, visualizing the consequences of not completing all planned tasks.

5.1 Project Management Concept

While the focus of our concept lies on gamification, a gamified tool also needs to provide helpful features to the users, otherwise users have little reason to use it actively. We aim to provide students with a tool that is useful to their software development process, even without any gamification features. As there already exist many Issue Management Systems (IMS), a strong emphasis is on features not offered by classic IMSs like *GitHub Projects* or *Atlassian JIRA*. A typical IMS feature is a *Kanban Board*, which shows the issues in columns representing their working state. *DinoDev* also provides such a board but extends it with warnings when users try to make unusual state changes, e.g., moving a completed issue back in the backlog, and a dialog that reminds the user of a customizable *Definition of Done* when closing an issue. This is based on our observation that the *Defintion of Done* is sometimes neglected in student projects. Furthermore, *DinoDev* provides a global activity overview (Figure 4), showing what is happening in the project. Many established IMSs provide a log of events for each issue, i.e., the system shows any change in the issue data in a timeline, such as changing the description or status. However, a joint overview of the activity across all issues is helpful. For example, students might only notice someone commenting on an assigned issue if they open the issue overview and check for new timeline items. While IMSs often mitigate this problem with email notifications, a global overview of all the events in a project is still a helpful feature, especially for the Product Owner and Scrum Master to assess the progress in the sprint. *DinoDev* provides this view on its main page, as shown in Figure 4. The events shown are filtered to show only the most relevant information, including created issues, completed issues, and opened and merged pull requests. Further project management features include visualizing selected statistics such as the sprint velocity, creating and commenting on issues, and the assistance in Scrum events as discussed in Section 5.4.

To avoid reimplementing a new IMS from the ground up, our concept incorporates data from external systems to provide the necessary data for the tool. Thus, *DinoDev* requires issue data from an existing IMS and pull request data from a Code Repository System (CRS) like *GitHub*, even though the latter is only used for the implementation of game elements for now. These external systems allow students to be delegated to these systems for more advanced operations. Thus, *DinoDev* is designed to work alongside an established IMS and CRS, combining the advantages of using industry-standard software with the custom functionality of the tool, albeit with the occasional need for users to switch between systems. This way, we avoid redundancy and keep the implementation effort manageable. The connection to the external systems is done via adapters, which translate the data to the *DinoDev* model and vice versa. Adapters allow the extension of the tool to work with other systems that provide an API, thus making the tool more versatile.

5.2 Team-Oriented Gamification

The team-oriented gamification concept aims to motivate the team as a whole to reach the sprint goal, work together successfully and productively, and support each other. Including various team-focused gamification elements addresses the supervisors' concerns regarding gamification creating distractions or fostering excessive competitiveness. Thereby, the following areas were considered:

5.2.1 Dinosaur Park. The dinosaur park is intended to create incentives to achieve the sprint goal and increase sprint velocity. Instead of a zoo with virtual animals, we decided on a dinosaur park, along with the related dinosaur-themed learning platform IT-REX for computer science students [56]. At the beginning of each sprint, the team can vote on the species of the dinosaur they want to hatch as a sprint mascot. Only a few species are available initially, but the team can unlock new species by increasing their sprint velocity. Additionally, the team can decide on a name for the dinosaur. We intend to create an emotional connection between the team and the dinosaur by letting the team name it, thus making the threat of the dinosaur leaving the park more effective, which should motivate the team to complete all the planned tasks of the sprint. The dinosaur park visualizes team progress in a non-intrusive way, aligning motivation with achieving sprint goals without directly pressuring individual students. As soon as the sprint starts, the dinosaur is shown on the main page of the project in *DinoDev*, as seen in Figure 4, next to the student activity in the project.

5.2.2 Social Engagements. We use multiple interactive elements that encourage social interactions between the team members. First, it is possible to post messages in the global activity overview, which are visible to all team members (Figure 4). Students can respond to these global messages. The same applies to messages on issues. These get additionally synchronized with the underlying IMS to ensure that all team members are informed about the discussion. Finally, students can react to any global and issue-specific activity with a heart emoji. This is similar to the like feature on social media platforms and is a simple way for students to show appreciation for the contributions of other team members, encourage them, or show

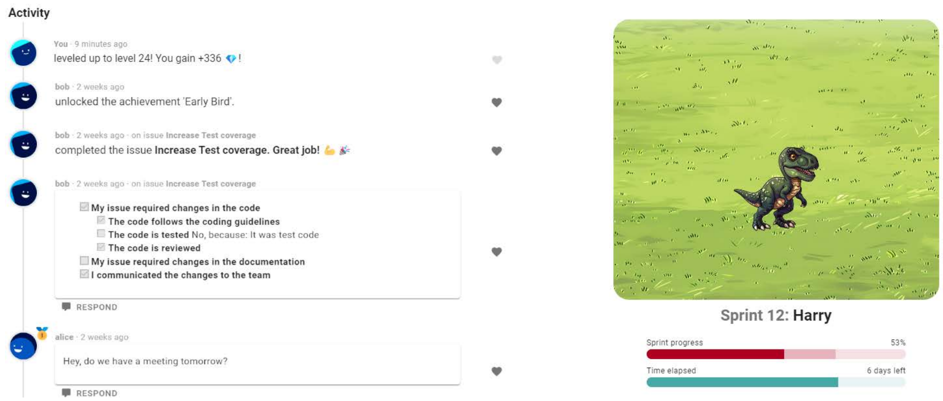

Figure 4: Sprint overview with student activity and sprint mascot "Harry".

that they agree with a message. These social engagement features encourage communication and support without interfering with task completion.

5.2.3 Reminder and Progress Bars. The reminder and progress bars are designed to create transparency, motivate students to reach their sprint goal, and alert them if they run late. Also, if two or fewer days are left in the sprint, a reminder is shown once a day in the global activity overview. It informs the students about the upcoming end of the sprint and states how many story points (SPs) are still open. Otherwise, a reminder is shown every two days, and only if the team is behind schedule, i.e., the percentage of completed SPs is lower than the percentage of the sprint duration that has passed.

The progress bar is placed below the sprint mascot (Figure 4) on the main page of *DinoDev*, which shows the progress of the sprint. It changes color depending on the progress of the sprint: (1) The progress bar is green, if the percentage of completed tasks is higher than the percentage of the sprint duration that has passed, meaning the team is ahead of schedule. (2) The progress bar is yellow, if the team is slightly (up to 10%) behind schedule. (3) Otherwise, the progress bar is red. The lighter part of the progress bar represents how many tasks are not completed yet, but in progress. Figure 4 shows an example where the team is behind schedule.

5.2.4 Gold Challenge and Unlockables. The gold challenge is a form of objective that incentivizes an increase in sprint velocity through rewards for raising the sprint workload. If the previous sprint was successful, the gold challenge is achieved by completing more than the previously planned SPs, i.e., by increasing the sprint velocity. In the case that the previous sprint was not successful, i.e., not all planned tasks were completed, it is earned by completing more SPs than the previously completed SPs. The first case ensures that *DinoDev* does not discourage the team from completing extra tasks. If the challenge were always based on the completed SPs, the team would be punished for completing more SPs than planned. The second case ensures the team is not overly punished for a failed sprint. If the challenge were always based on the planned SPs, a failed sprint would make it hard to achieve the gold challenge in the next sprint, as the team would need to complete even more SPs than they failed to do in their previous sprint. Figure 5 highlights how the gold challenge is presented in the sprint goal planning. In this example, two more SPs are required to achieve it, as eight SPs were completed in the previous sprint, and only seven are planned for the current sprint.

5.3 Student-Oriented Gamification

The student-oriented gamification concept is intended to promote the motivation of individual students to actively participate in the project, to fulfill their tasks and duties, and to increase their contribution to the project. Thereby, the following areas were considered:

5.3.1 Achievements. *DinoDev* uses achievements to encourage students to take specific actions. Some of the achievements are simple progress achievements that are unlocked by completing a certain number of issues, others are more specific, e.g., to incentivize students to contribute early in the sprint. The student profile shows the achievements attained and the progress towards those that still need to be unlocked. Thus, students can see their achievements and those of other students.

5.3.2 Experience Points (XP). We use an XP-based progression system to reward students for various actions. The requirements analysis revealed that the SPs of issues are a meaningful metric for the progression system. Thus, completing issues rewards XP, 1000 XP per SP of the issue. The XP are distributed evenly among the assignees of the issue. Although the contribution of each assignee might be unevenly distributed, we decided to distribute the XP evenly to avoid conflicts between team members and to keep the system simple. Coding is an essential part of software development; thus, creating and reviewing pull requests also rewards XP. However, this XP gain is significantly lower than for completing issues, as otherwise, we would reward all coding-related actions twice: once for coding and once for completing the issue. In addition, participating in meetings awards XP to acknowledge the time spent in the meeting. The meeting leader, the person in the team guiding the meeting, receives five times the XP of a regular participant. The reason behind this is that the meeting leader is usually the Scrum Master, who does not contribute to the development of the product by working on issues and, therefore, does not receive XP

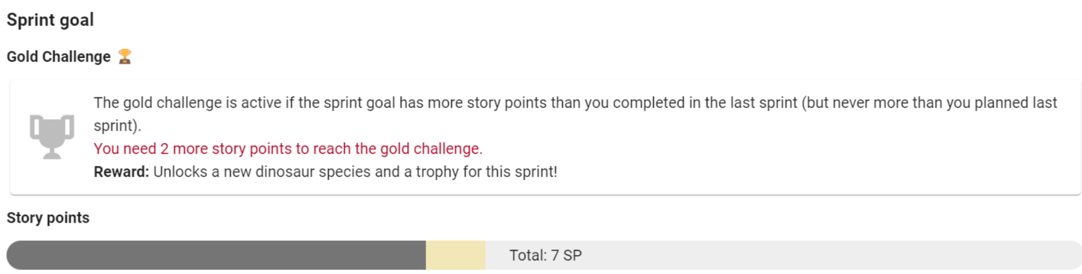

Figure 5: The gold challenge in the sprint goal planning.

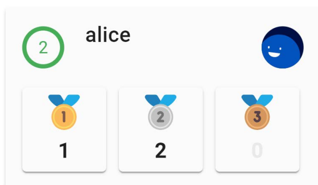

Figure 6: Student's level and medals in a project.

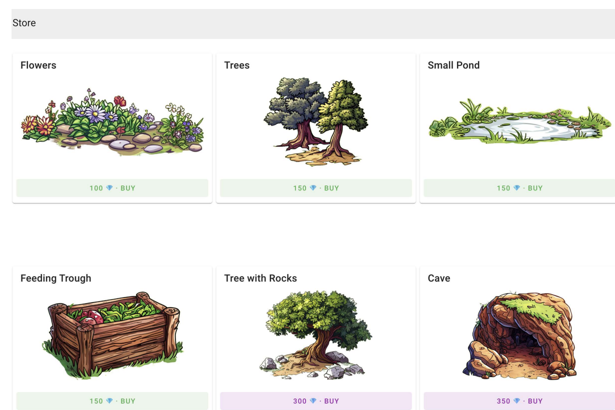

Figure 7: Shop to buy decorations with virtual currency.

for completing issues. The Product Owner role is currently only considered to a small extent by awarding XP for creating issues.

5.3.3 Level System. The level system aims to motivate students to level up through the project processes and thus accomplish their tasks and duties. After gaining a certain amount of XP, the student levels up. The level represents the student's progress and is displayed in the sidebar for the students themselves and in an overlay, as displayed in Figure 6, for all students. Displaying the student's level to the team is a way to acknowledge the student's contributions and to motivate other team members to catch up if their level is lower.

The amount of XP required for a level-up increases with each level. An increase in the required XP for each level-up is a way to incentivize the students to increase their productivity over time since it becomes harder to level up the higher the student's level is.

5.3.4 Virtual Currency and Decorations. The reward system of *DinoDev* focuses on the player type "Player", who is motivated by extrinsic rewards and is based on a virtual currency that is used to purchase decorations for the dinosaur park. This currency is earned by leveling up. As the difficulty of leveling up increases with each level, the students earn more currency the higher their level is. At the beginning of the sprint, the large green area around the dinosaur is empty (left image in Figure 3). This area represents the "enclosure" of the dinosaur. The students can fill it with decorations, which they can purchase in a shop using virtual currency. This shop is presented in Figure 7. The decorations are visible to all team members and are a way to customize the tool. The dinosaur area filled with decorations is displayed in the right image in Figure 3. Besides that, the decoration elements have no impact on other game elements.

5.3.5 Medals. *DinoDev* awards a medal at the end of each sprint to the three students with the greatest contribution to encourage ambition to perform strongly. The three top contributors measured by SPs completed in the sprint receive a bronze, silver, or gold medal. In case of a tie, the winner is determined randomly. Winning a medal has further implications: (1) The student receives a bonus of 50, 75, or 100 virtual currency, respectively. (2) The student receives a badge of the respective medal shown above their avatar, for example, in the global activity overview (e.g., the student Alice in Figure 4). (3) The student profile displays the total number of medals won, as shown in Figure 6. The intention behind the medals is to reward the top contributors and to create a subtle competition among the team members. This competitive element is deliberately limited to the retrospective meeting and remains hidden during the sprint to address the supervisors' concerns about introducing undue competitiveness among students. In showing only the top contributors, we aim to avoid demotivating team members who are not among them. Thus, we avoid the disadvantages of a leaderboard, where the lower-ranked students might feel demotivated. However, for small student teams, the medal gamification element might be demotivating if the majority of team members are awarded except one or two students.

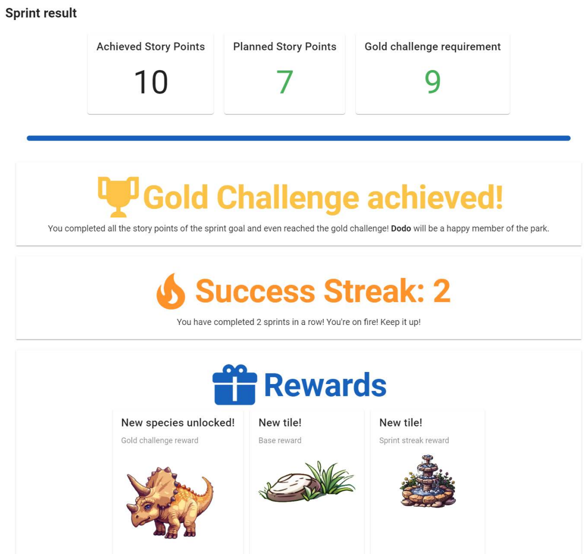

Figure 8: Rewards for a successful sprint.

5.4 Scrum Events

Some game elements are already commonly used in Scrum: "Planning Poker", for example, is a gamified way of issue estimation. Furthermore, various suggestions to gamify retrospectives exist, such as "The Sailboat Retrospective", "Mad Sad Glad" or "The 4Ls" [22]. We combine these existing approaches with additional game elements. *DinoDev* includes a simplified planning poker feature using T-Shirt sizes (XS-XL) to estimate issues, which are mapped to SPs. In addition, we use voting mechanisms to let the team decide on the dinosaur's species and name for the next sprint mascot. While the species that can be voted on are predefined, the team can freely suggest names for the dinosaur. The two additional voting mechanisms may slightly increase the time needed for the planning meeting, but they are intended to be completed quickly and should not significantly impact the duration.

In the sprint retrospective, the rewards for a successful sprint are distributed to the team. This screen is shown in Figure 8 and combines the gamification elements of the gold challenges and unlockables, as well as rewarding a streak of the student team if they achieve the sprint goal in a series of successive sprints. The medals of a sprint are also awarded to the students in the retrospective. *DinoDev* also supports the team in their daily standup meetings with helpful features for keeping time and discussing issues, but these do not contain any further gamification elements.

5.5 Considered Player Types

The concept of *DinoDev* is designed to consider all six player types by Marczewski [30]. While addressing some player types is more straightforward, others require more thought. The most challenging player type to address is the *Disruptor*, which is also the least common player type in our user analysis. We aim to address all of them to some extent:

Achievers are motivated by completing challenges and achieving goals. We aim to address them by providing the gold challenge,

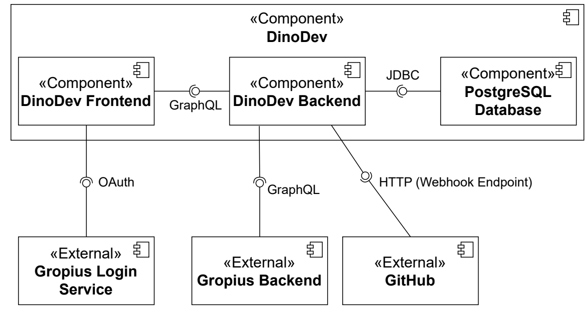

Figure 9: Architecture of *DinoDev*.

achievements, and the leveling system. Furthermore, the medals and unlocking new species and decorations are also intended to motivate *Achievers*.

Socializers are motivated by social interactions and relatedness. We address this player type by providing the global activity overview, where users can post messages and like activities. The elements of the planning and retrospective meetings, such as voting and retrospective games, are intended to increase social interactions.

Free Spirits are motivated by autonomy and creativity. The possibility of suggesting names for the dinosaur and the freedom to choose decorations for the dinosaur park are ways to address this player type. Unlocking new content is also a way for *Free Spirits* to explore the tool.

Philanthropists are motivated by helping others. The global activity overview provides a way to help others and share knowledge. If they take on the role of the meeting leader, they have an administrative role guiding the meeting. According to Krath and von Korflesch [26], challenges can also be appropriate for *Philanthropists*, which the gold challenge addresses.

Players are motivated by extrinsic rewards. The virtual currency, the unlocking of new species and decorations, the progression system, and the medals are the extrinsic rewards in *DinoDev*, which are intended to motivate *Players*.

Disruptors are motivated by changing the status quo. The voting mechanisms in the planning and the possibility of proposing names for the dinosaur are opportunities for *Disruptors* to drive change. Furthermore, the placement of decorations in the dinosaur park does not follow strict rules. They can be placed freely, allowing for creative or nonsensical arrangements.

6 DinoDev Prototype

DinoDev was implemented as a client-server application with a web interface, a single backend server, and a database. Figure 9 provides an overview of the tool's architecture, which is inspired by the proposed architecture for SE gamification of Pedreira et al.[43]. The code of *DinoDev* is open-source and publicly available on GitHub[3].

The tool uses an external IMS and CRS with which the backend interacts via adapters. As the CRS, GitHub is used due to its high popularity. *DinoDev* only reacts passively to the events in GitHub, using a webhook. The IMS used in the prototype is *Gropius*[4], a

[3] https://github.com/MEITREX/dinodev
[4] https://github.com/ccims

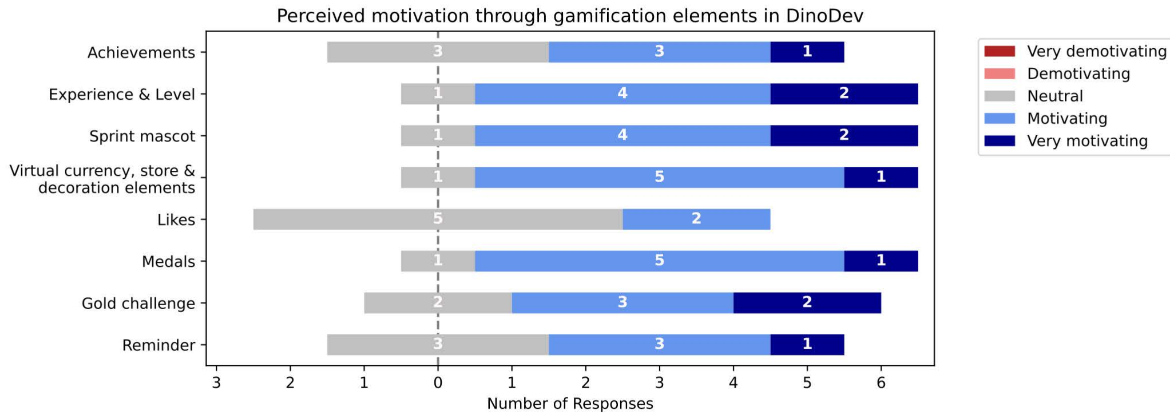

Figure 10: Perceived motivation through gamification elements in *DinoDev*.

cross-component issue management system that incorporates system architecture into the issue management process to enable better tracking and management of issues across distributed software components [55, 57, 59]. We chose Gropius due to the synchronization capabilities of its issue data with other popular IMSs such as *Atlassian JIRA*. With these capabilities, Gropius provides a unified way to access these other IMSs and solves technical hurdles such as API rate limits. Furthermore, it is used for authentication, which simplifies the implementation and allows the students to reuse their Gropius account.

7 Evaluation

This chapter outlines the evaluation of the *DinoDev* concept. For this purpose, the implemented prototype was used in a real student software development project at the University of Stuttgart. The following sections describe the evaluation process (Section 7.1), report the results (Section 7.2), and finally discuss the results (Section 7.3).

7.1 Evaluation Process

We evaluated *DinoDev* in the context of one student software development project containing seven students in their Bachelor's degree, which took place in the summer semester of 2024. *DinoDev* was introduced in the third sprint of their project, although the first sprint was mainly used to familiarize themselves with the project and the existing code base. Following the introduction of *DinoDev*, the students used the tool for two sprints, corresponding to four weeks. Before, during, and after the study, the students completed questionnaires: (1) A pre-study questionnaire that the students completed before using *DinoDev*, containing questions about the agile methodology used and students' motivation. (2) A mid-study questionnaire after two weeks (one sprint) about the usefulness of features and how particular gamification elements influenced the students. (3) A post-study questionnaire that the students completed after four weeks (two sprints) of using *DinoDev* to gain insight into the impact on student motivation and team productivity. The full questionnaires and the result data can be accessed on Zenodo[2].

7.2 Results

Regarding the statement that "using *DinoDev* increased my motivation to work on tasks", all participants except one agreed. Moreover, the same number of participants stated that the project management components of *DinoDev* increased their motivation. The gamification features of *DinoDev* had a slightly smaller impact on the motivation, as one participant disagreed with the statement, that "the gamification features of *DinoDev* increased my motivation to work on tasks" and two were neutral. The other four respondents agreed that the gamification features increased their motivation, while two strongly agreed. The results regarding the effect of specific gamification elements in *DinoDev* are shown in Figure 10. None of the participants perceived any of the gamification elements as demotivating. The most motivating elements were experience and level, the sprint mascot, medals, and the decoration elements, which can be purchased with virtual currency. All participants except one rated these elements as motivating or very motivating. The least motivating elements were the likes, which refers to the feature that allows users to like activities in the global activity feed. Two participants rated this feature as motivating, while the other five were neutral. All results can be accessed on Zenodo[2].

Sprint velocity is a standard metric in Scrum that measures a team's productivity. To obtain the sprint velocity, we refer "Sprint A" to the sprint before the study, "Sprint B" to the first, and "Sprint C" to the second sprint of the study. We observe a slight increase of four SPs in the planned story points from Sprint A to Sprint B and a slight increase of two SPs in the achieved SPs from Sprint A to Sprint C. In Sprint B, the team only completed 49 SPs instead of the planned 64 SPs, so the sprint goal was not met. The following Sprint C was a success again, where the team completed 66 SPs, exceeding the sprint goal by two SPs.

7.3 Discussion

The introduction of *DinoDev* had a slight impact on student motivation. The results from the questionnaires indicate that *DinoDev* generally increased students' motivation to work on tasks, especially among those students who were not already motivated. All participants except one agreed that *DinoDev* increased their motivation to complete tasks. Four of the seven participants agreed that the gamification features increased their motivation, and a correlation analysis suggests that the game concept is more motivating for the *Player* and *Achiever* types. However, we assume the overall effect on motivation to be relatively small, as the analysis of factors that (de-)motivate students only revealed aspects related to the project itself. This indicates that addressing the project-specific factors that demotivate students is crucial to avoid a lack of motivation.

The impact of *DinoDev* on productivity, measured through sprint velocity, presents mixed results. While there was a slight increase in completed story points in the final sprint of the study, the second sprint was less successful, where the team did not meet the sprint goal. The failure of the second sprint was attributed to underestimating the complexity of the tasks and adding new tasks during the sprint, not to the introduction of *DinoDev*.

8 Threats to Validity

Internal validity: The evaluation only looks at seven students from a joint project. It is possible that the results are influenced more by individual or group dynamics than by the *DinoDev* tool itself. Students' self-reported motivation may also be biased or inaccurate, as their perceptions could be influenced by social desirability or different interpretations of the questionnaire items. In addition, the novelty effect of using a gamified tool may have temporarily increased engagement, which may not reflect long-term motivation. Further studies, especially at other universities, and over longer periods of time should therefore be conducted in the future.

External validity: The small sample size and the context of a single project limit the generalizability of the results. However, as our project is similar to projects in other universities, the results should, in principle, be generalizable, potentially to varying degrees. Likewise, the specific integration of tools such as Gropius and GitHub may not match workflows in other environments, affecting the transferability of results to broader SE education contexts. However, GitHub is an established tool, particularly in academic environments, and the use of Gropius as a form of middleware can be added to any project, so we consider this risk to be negligible.

Construct validity: The use of questionnaires to measure students' motivation cannot fully capture the complexity of their engagement with the Scrum process and *DinoDev*. In particular, self-assessment may not always reflect actual behavior, and external factors (e.g., project workload, project topic, or influence of fellow students) could influence responses. However, student perceptions of the functionality and usability of *DinoDev* to promote motivation could be examined and evaluated using this evaluation method.

Conclusion Validity: With only two sprints (four weeks) after the introduction of *DinoDev*, the duration of the evaluation was rather limited, making it difficult to assess longer-term effects on motivation through the tool. Further studies over longer periods of time should, therefore, be conducted in the future.

9 Lessons Learned

We gathered some lessons learned that provide insights into the use of gamification in agile student software development projects. They are valuable for advancing research and gamification in PBL and enhancing student motivation in SE education. The lessons learned are also useful for lecturers who want to use gamification to incentivize students to work on their group projects. The key insights comprise: (1) Students particularly appreciated *DinoDev*'s guidance through Scrum events by providing gamified and visually appealing features that encourage collaboration within the student team and make each student's work and contributions visible. Nevertheless, its acceptance highly depends on the individual students who use *DinoDev*. (2) Gamified elements must be explicitly linked to the learning objectives. The gamified and appealingly designed Scrum event support features in *DinoDev* were appreciated by the students, along with the incentives to achieve the sprint goal. Designing gamification mechanics that directly reinforce concepts like agile practices, Scrum roles, and project management skills is crucial for maximizing their educational impact. (3) *DinoDev*'s gamification design predominantly benefited certain player types, such as "Achievers" and "Players", while engaging others, like "Socializers" less effectively. This highlights the need for iterative design informed by diverse student feedback to create a more inclusive gamified experience. Incorporating features that cater to various motivations, such as exploratory tasks or collaboration-focused rewards, can broaden the tool's appeal and effectiveness.

10 Conclusion and Future Work

This paper introduces an approach for integrating gamification into project-based learning in Software Engineering education. The goal was to increase student motivation to contribute to software development projects, enhance team productivity, and offer tailored project management features for Scrum-based processes in an educational context. An anonymous online questionnaire involving 27 students and seven supervisors who previously participated in such projects revealed that Scrum, with minor variations, is commonly used and identified key reasons for motivational challenges in student projects.

Based on these insights, we developed and implemented a gamified project management tool, *DinoDev*, combining project management with gamification elements. The *DinoDev* prototype was deployed in a student project with seven students at the University of Stuttgart. Results showed that the objective of increasing student motivation (*RQ1*) was achieved, though the effect was subtle. However, results regarding the impact of gamification on team productivity (*RQ2*) were inconclusive, largely due to inaccurate effort estimation and lack of experience, which affected sprint velocity. A slight increase in sprint velocity was observed after introducing *DinoDev*, but this could not be conclusively attributed to the tool. Future studies are needed to provide more definitive conclusions.

Despite these limitations, the study's results are promising, indicating that gamification in PBL for SE education can be beneficial. *DinoDev* mechanisms promote task completion and improved sprint velocity, potentially leading to more successful projects, benefiting both students and supervisors. The gamification concept, while particularly appealing to developers, needs refinement to address roles like Product Owner and Scrum Master.

Future work will evaluate the concept with larger and more diverse samples, possibly including projects from other universities. Further iterations using the GOAL methodology will refine the gamification elements and better assess the tool's impact and effectiveness.

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